

KHALYASAVEY, Russian FederationWMO: **236560**Lat: **63.384N**Lon: **78.324E**Elev: **67**StdP: **100.52**Time Zone: **5.00 (E05)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.4	(c) -41.4	(d) -48.5	(e) 0.0	(f) -45.4	(g) -45.0	(h) 0.0	(i) -41.5	(j) 6.6	(k) -15.3	(l) 6.3	(m) -15.1	(n) 1.2	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.3	18.2	27.3	17.6	24.8	16.8	19.9	25.4	18.9	24.6	17.8	23.4	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.9	22.1	16.8	12.1	21.3	15.5	11.1	20.2	57.2	25.5	54.1	24.1	50.5	23.7	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.3	(b) 6.4	(c) 5.7	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(4) 7.3	(5) 6.4	(6) 5.7	(7) DB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A
			(21) WB	(22) -46.6	(23) 20.4	(24) 3.1	(25) 1.8	(26) -48.8	(27) 21.7	(28) -50.6	(29) 22.7	(30) -52.3	(31) 23.7	(32) -54.6	(33) 25.0	(34) 25.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-23.9	-20.0	-10.6	-4.0	2.2	14.5	17.6	13.2	6.5	-3.1	-16.0	-18.3
	DBStd	15.55	9.76	9.54	8.16	6.06	5.62	5.65	4.24	3.64	4.38	5.55	9.03	9.80
	HDD10.0	5421	1051	841	639	421	252	23	1	10	123	406	779	877
	HDD18.3	8006	1309	1074	897	671	502	139	67	165	354	663	1029	1135
	CDD10.0	531	0	0	0	0	9	159	236	108	18	1	0	0
	CDD18.3	75	0	0	0	0	1	25	44	4	0	0	0	0
	CDH23.3	804	0	0	0	0	12	257	502	33	0	0	0	0
	CDH26.7	234	0	0	0	0	3	74	153	4	0	0	0	0
	WSAvg	2.8	2.5	2.6	3.1	3.2	3.4	3.0	2.8	2.6	2.6	2.8	2.6	2.6
	PrecAvg	511	23	19	24	29	38	52	60	79	54	56	40	33
(15) Precipitation	PrecMax	652	44	36	47	58	72	108	146	138	87	107	80	65
	PrecMin	386	3	7	3	7	17	6	10	32	14	26	16	13
	PrecStd	81	11	7	12	13	13	23	32	26	18	22	16	12
	PrecStd	81	11	7	12	13	13	23	32	26	18	22	16	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	-4.2	1.3	4.2	11.0	24.2	31.5	32.2	26.6	20.1	13.1	3.0	-0.3	-0.3
	0.4% MCWB	-4.8	0.7	1.0	5.5	14.8	18.0	19.2	18.1	14.7	8.1	2.7	-1.0	-1.0
	2% DB	-6.7	-2.6	2.0	8.3	18.8	28.8	30.2	23.9	17.4	7.5	0.3	-1.8	-1.8
	2% MCWB	-7.2	-3.3	-0.3	3.6	12.1	18.0	17.9	16.8	13.1	5.9	-0.4	-2.6	-2.6
	5% DB	-8.4	-4.7	1.0	6.3	14.2	26.5	28.4	21.8	15.2	5.6	-1.1	-3.8	-3.8
	5% MCWB	-9.0	-5.3	-0.5	2.7	9.4	17.1	18.2	15.9	11.5	3.8	-1.5	-4.3	-4.3
	10% DB	-10.4	-7.2	-0.2	4.5	10.9	23.7	26.3	19.6	13.3	3.2	-2.4	-5.8	-5.8
	10% MCWB	-10.9	-7.9	-1.6	1.8	7.0	16.4	17.3	14.8	10.5	1.9	-2.8	-6.4	-6.4
	0.4% WB	-4.4	0.7	1.4	5.9	15.2	20.9	20.9	20.2	15.3	8.5	2.4	-0.8	-0.8
	0.4% MCDB	-4.2	1.2	2.8	10.2	23.7	26.5	27.7	23.8	18.2	12.4	2.8	-0.3	-0.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-7.1	-3.2	0.3	4.1	12.5	19.5	20.1	18.2	13.5	6.0	-0.1	-2.6	-2.6
	2% MCDB	-6.7	-2.6	1.6	6.9	18.2	24.3	26.6	21.5	16.7	7.5	0.3	-1.9	-1.9
	5% WB	-8.9	-5.3	-0.5	3.1	9.7	18.4	19.3	16.6	12.2	4.0	-1.5	-4.4	-4.4
	5% MCDB	-8.4	-4.8	0.7	6.0	13.8	24.4	25.6	20.5	14.8	5.1	-1.1	-3.9	-3.9
	10% WB	-10.9	-7.8	-1.5	2.0	7.1	17.2	18.4	15.6	10.6	2.0	-3.1	-6.5	-6.5
	10% MCDB	-10.4	-7.2	-0.2	4.1	10.4	22.9	24.4	18.8	13.0	2.9	-2.6	-6.0	-6.0
(35) Mean Daily Temperature Range	MDBR	8.1	8.5	9.4	9.6	7.9	9.8	9.9	8.4	6.4	4.8	7.4	7.8	7.8
	5% DB MCDBR	9.5	7.8	8.5	10.5	14.0	13.3	13.8	11.9	9.2	5.5	5.3	8.1	8.1
	5% DB MCWBR	9.4	7.6	7.2	7.0	8.3	4.8	4.3	6.0	5.0	4.0	5.6	8.0	8.0
	5% WB MCDBR	9.4	8.1	8.0	9.5	13.7	11.5	11.5	9.8	8.4	5.3	5.5	8.3	8.3
	5% WB MCWBR	9.3	8.0	7.1	6.8	8.8	5.3	4.5	5.8	5.1	4.3	5.4	8.2	8.2
	5% WB MCWBR	9.3	8.0	7.1	6.8	8.8	5.3	4.5	5.8	5.1	4.3	5.4	8.2	8.2
(40) Clear-Sky Solar Irradiance	taub	0.232	0.282	0.332	0.353	0.365	0.381	0.430	0.402	0.349	0.304	0.226	0.226	0.226
	taud	1.912	2.019	1.983	2.042	2.255	2.397	2.232	2.294	2.399	2.214	1.940	1.879	1.879
	Ebn at Noon	342	616	729	811	844	839	776	764	740	597	340	140	140
	Edh at Noon	34	71	112	131	117	104	120	103	76	59	31	16	16
(44) All-Sky Solar Radiation	RadAvg	0.23	1.04	2.68	4.65	5.39	5.46	4.96	3.41	2.06	0.78	0.28	0.10	0.10
	RadStd	0.02	0.08	0.14	0.46	0.48	0.41	0.52	0.42	0.30	0.11	0.06	0.01	0.01

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+1.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page